

MID-TERM EXAMINATION

Subject: python

Paper: Set A | Duration: 30 Minutes | Max Marks: 50 | Date: 22-02-2026

Total Questions: 1 | Answer all questions.

Section 1: MCQ (1 Questions - 1 Marks)

Q1. [1 marks] (Hard)

Consider the following Python code defining a custom descriptor and a class using it:

```
class ValueTracker:
    def __init__(self, initial_value):
        self._value = initial_value
        self.name = None

    def __set_name__(self, owner, name):
        self.name = name

    def __get__(self, instance, owner):
        if instance is None:
            # Access via class
            return f"Class-level default for {self.name}: {self._value}"
        else:
            # Access via instance
            if self.name in instance.__dict__:
                return f"Instance-specific value for {self.name}: {instance.__dict__[self.name]}"
            else:
                return f"Descriptor's initial value for {self.name}: {self._value}"

    def __set__(self, instance, value):
        # This method's presence makes ValueTracker a data descriptor
        instance.__dict__[self.name] = value

class MyClass:
    data = ValueTracker("default")

# Execution sequence
obj = MyClass()
output = []
output.append(MyClass.data)
output.append(obj.data)
obj.data = "custom"
output.append(obj.data)
output.append(MyClass.data)
```

```
print("\n".join(output))
```

Which of the following sequences accurately represents the printed output?

A)

Class-level default for data: default

Instance-specific value for data: default

Instance-specific value for data: custom

Class-level default for data: custom

B)

Class-level default for data: default

Descriptor's initial value for data: default

Instance-specific value for data: custom

Class-level default for data: default

C)

Class-level default for data: default

Descriptor's initial value for data: default

Descriptor's initial value for data: custom

Class-level default for data: default

D)

Class-level default for data: default

Instance-specific value for data: default

Instance-specific value for data: custom

Descriptor's initial value for data: default

--- End of Question Paper ---